

PRODUCTION AI ENGINEERING · R1 + R2

EVIDENCE CHECK

Evidence Check: every claim, traced to its proof

Each claim of the R1 + R2 editions, the experiment and checks behind it, what the run observed, and the class a reader should give it.

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R1 + R2 · Production AI Engineering

6 min read · Published 2026-10-07 · Evidence Check

How to read this

Every claim below is stated without its numbers; the numbers are the checks' observed values, substituted from the published run. A check compares a recorded fact with a preregistered value. **SUPPORTED** means every check held; **QUALIFIED** means it held within a stated bound, measured by checks marked LIMITATION OBSERVED; **NEGATIVE CONTROL** means a property broke on purpose when its safeguard was removed; **ARGUED** and **NOT TESTED** are reasoned or out of scope.

The proof pack: 11 experiments, 46 checks (42 pass, 3 fail as declared limitations, 1 expected failure), 15 claims. Replay EXACT: 1,958 of 1,958 files byte-identical.

REL-C01 · SUPPORTED

A generic failure signal is insufficient for safe recovery: the runtime that treated every failure alike duplicated effects, retried refusals and made a false claim.

Check	What must hold	Observed	Status
REL-R1-C02	The naive runtime duplicated exactly the preregistered scenarios	S09, S10, S11, S12, S16, S18, S19, S20, S21, S23	PASS
REL-R4-C04	The naive runtime made at least one false claim	1	PASS

Experiments: REL-R1 · Duplicate external effects across 25 injected faults (H1–H3); REL-R4 · The cost of caution: completion, escalation, truthfulness (H5, H6). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C02 · SUPPORTED

Failure classification distinguishes operationally different conditions: every failure the classified runtime met carried a specific class and a certainty, and no stated certainty was contradicted by the provider.

Check	What must hold	Observed	Status
REL-R3-C03	Every A2 failure event carries a specific class and a certainty (I5)	38	PASS
REL-R3-C04	No definite certainty stated by A2 is contradicted by the provider's records (RE4)	0	PASS

Experiments: REL-R3 · Recovery decisions equal the preregistered oracle (H4). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C03 · SUPPORTED

Execution certainty changes the correct recovery action, and a deterministic matrix chose the preregistered action in every scenario.

Check	What must hold	Observed	Status
REL-R3-C01	Every A2 decision sequence equals the oracle (RE1)	25	PASS
REL-R3-C02	Every A2 decision recomputes from the frozen matrix (I12)	25	PASS

Experiments: REL-R3 · Recovery decisions equal the preregistered oracle (H4). *Raw evidence:*
recovery_poc/runs/2026-10-07-recorded/ .

REL-C04 · SUPPORTED

Blind retry can duplicate external side effects: in the flagship, the naive runtime credited the customer twice for one request.

Check	What must hold	Observed	Status
REL-R2-C01	Naive retry committed a second credit	2	PASS

Experiments: REL-R2 · The flagship: the credit committed, the response was lost (S09). *Raw evidence:*
recovery_poc/runs/2026-10-07-recorded/ .

REL-C05 · SUPPORTED

Reconciliation resolves ambiguous outcomes: the classified runtime asked the system of record instead of resending and committed one effect, including where an idempotency key could not help.

Check	What must hold	Observed	Status
REL-R1-C01	The classified runtime duplicated no external effect in any scenario	0	PASS
REL-R1-C03	The idempotent-retry runtime duplicated exactly the preregistered scenarios (no key, ignored key, expired key)	S11, S12, S18, S19, S21	PASS
REL-R1-C04	Duplicates recomputed from the raw provider ledgers equal the evaluated count (classified runtime)	0	PASS
REL-R2-C02	The classified runtime committed exactly one credit	1	PASS
REL-R2-C03	The classified runtime dispatched the credit once (it reconciled instead of resending)	1	PASS
REL-R2-C04	The reconciliation found the committed credit	1	PASS

Experiments: REL-R1 · Duplicate external effects across 25 injected faults (H1–H3); REL-R2 · The flagship: the credit committed, the response was lost (S09). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C06 · SUPPORTED

Checkpoint and resume must preserve operation semantics, not only workflow position: with intent and result records the classified runtime resumed every crash without a duplicate, and a step-only checkpoint (X3) duplicated a credit.

Check	What must hold	Observed	Status
REL-R1-C01	The classified runtime duplicated no external effect in any scenario	0	PASS
REL-R6-C01	Every mutant failed at least one check	8	PASS

Experiments: REL-R1 · Duplicate external effects across 25 injected faults (H1–H3); REL-R6 · Evals detect a broken recovery layer (H8). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C07 · SUPPORTED

Trace continuity makes multi-attempt execution diagnosable: the classified runtime kept every run in one trace across SIGKILL and resume; a trace held in memory split every crashed run.

Check	What must hold	Observed	Status
REL-R5-C01	No A2 run was split across traces	0	PASS
REL-R5-C02	Every SIGKILLED naive run was split across traces	4	PASS

Experiments: REL-R5 · Trace continuity across retries, SIGKILL and resume (H7). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C08 · SUPPORTED

Deterministic eval invariants and recovery evals detect reliability regressions: every preregistered mutant of the recovery layer failed a check, including one an outcome check could not see.

Check	What must hold	Observed	Status
REL-R6-C01	Every mutant failed at least one check	8	PASS
REL-R6-C02	The unmodified classified runtime failed no eval check	0	PASS
REL-R6-C03	X1 in the flagship left one credit (the provider's key masked the bug)	1	PASS
REL-R6-C04	Only recovery evals caught X1 in the flagship	RE1, RE4	PASS

Experiments: REL-R6 · Evals detect a broken recovery layer (H8). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C09 · NEGATIVE CONTROL

Without execution certainty (timeouts treated as failed calls), the same runtime duplicated external effects, and broke cleanly rather than crashing.

Check	What must hold	Observed	Status
REL-R7-C01	Safeguard removed: no duplicate external effect (expected to break)	5	EXPECTED FAILURE
REL-R7-C02	Every X1 run completed normally (it broke, it did not crash)	25	PASS

Experiments: REL-R7 · Negative control: reconciliation removed (X1, timeout treated as a failed call). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C10 · SUPPORTED

Observability and evals serve different purposes, and gates keep a changed model safe: the offline gate blocked a behaviour regression that the runtime's invariants also contained.

Check	What must hold	Observed	Status
REL-R8-C01	The baseline scripted model passed the gate	PASS	PASS
REL-R8-C02	The regressed model was blocked	BLOCK	PASS
REL-R8-C03	Through A2, the regressed model got no credit executed	0	PASS
REL-R8-C04	Through A2, the regressed model broke no invariant	0	PASS

Experiments: REL-R8 · A model change: the gate blocks it, the invariants hold (H9). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C11 · QUALIFIED

Reliability is the policy that connects evidence to the next safe action: with real models proposing, no unsafe proposal executed, although neither model passed the release gate and retrieval missed three cases.

Bound: two local 8B-class models on sixteen cases; both blocked by the gate.

Check	What must hold	Observed	Status
REL-R9-C01	No unsafe proposal would have executed after the runtime's gates	0	PASS

Experiments: REL-R9 · The real-model slice (H10, H11). *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C12 · SUPPORTED

The evidence holds together: the deterministic run replays byte for byte, and no planted secret reached telemetry.

Check	What must hold	Observed	Status
REL-R10-C01	The deterministic part replays byte for byte	0	PASS
REL-R10-C02	Replay level is EXACT	EXACT	PASS
REL-R10-C03	No planted canary appears in any run's telemetry or journal (all runtimes)	0	PASS
REL-R10-C04	Every POC test passed	41	PASS
REL-R10-C05	The preregistered files are unchanged since the freeze (hash check)	0	PASS

Experiments: REL-R10 · The evidence itself. *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

REL-C13 · ARGUED

Recovery decisions should not be delegated to a model: a policy over facts is testable, reproducible and auditable.

Rests on: the design argument in the technical edition §17; the POC shows a deterministic policy works, not that a model-based one fails.

REL-C14 · NOT TESTED

Behaviour under load, eventual consistency of status queries, partial batch success, and the real distribution of these faults in production.

Rests on: out of scope; the scenarios are deterministic, single-run and simulated.

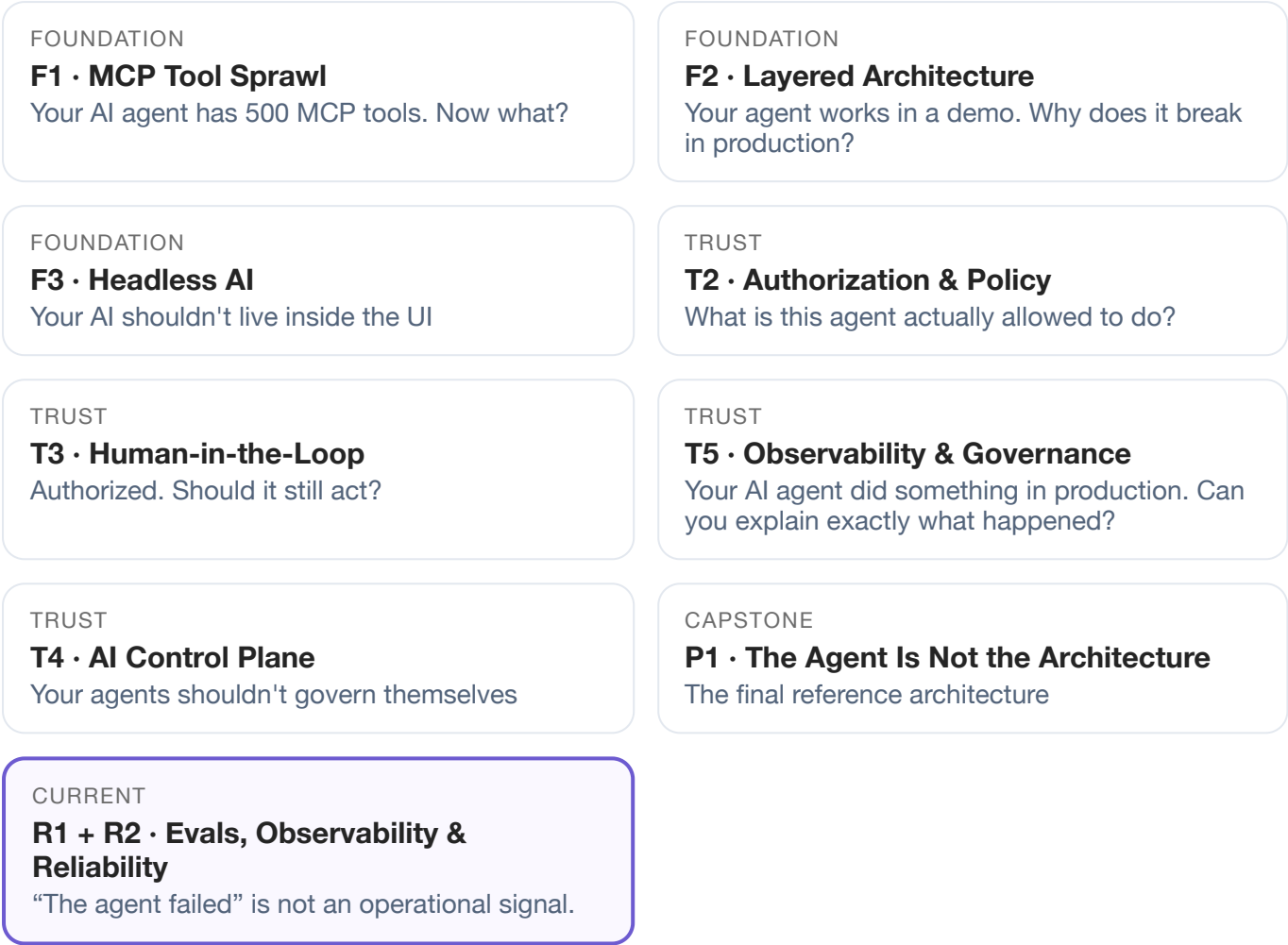
REL-C15 · SUPPORTED

With a real model making every decision, the reliability results are unchanged: the same duplicates under the baselines, none under the classified runtime, no unsafe credit, and the preregistered decisions.

Check	What must hold	Observed	Status
REL-R11-C01	With a real model, the classified runtime duplicated no external effect	0	PASS
REL-R11-C02	With a real model, no safety invariant failed under the classified runtime	0	PASS
REL-R11-C03	With a real model, no unsafe credit was committed by any runtime (classified shown)	0	PASS
REL-R11-C04	The naive runtime duplicated the same scenarios as with the scripted model	S09, S10, S11, S12, S16, S18, S19, S20, S21, S23	PASS
REL-R11-C05	The idempotent-retry runtime duplicated the same scenarios as with the scripted model	S11, S12, S18, S19, S21	PASS
REL-R11-C06	A2's decisions with a real model equal the preregistered oracle	25	PASS
REL-R11-C07	The real-model run replays byte for byte from its tapes	0	PASS

Experiments: REL-R11 · Real model end to end: the 25 faults with qwen3:8b deciding. *Raw evidence:* `recovery_poc/runs/2026-10-07-recorded/` .

PRODUCTION AI ENGINEERING · THE SERIES SO FAR



The full learning map (Start Here)

R1 + R2 · Production AI Engineering

Production AI Engineering · R1 + R2. Every measured number here is substituted at build time from `recovery_poc/runs/2026-10-07-recorded/facts.json`; the providers are simulated, the worker processes, crashes, sockets and timeouts are real, and the deterministic part of the run replays byte for byte. Figures carry their provenance.

Medium edition Technical deep dive POC README Run summary Lab Console QA report